



DAY 1 DPP (Daily Practice Problems) Introduction to AI/ML + Python Basics

This DPP is designed for complete beginners starting their journey in Artificial Intelligence, Machine Learning, and Python programming.

SECTION A — THEORY QUESTIONS

- Q1. Define Artificial Intelligence in your own words.
- Q2. What is Machine Learning?
- Q3. Differentiate between AI, ML, and Deep Learning.
- Q4. What is Supervised Learning? Give one real-life example.
- Q5. What is Unsupervised Learning? Give one real-life example.
- Q6. Explain Machine Learning using the teacher-student analogy.
- Q7. What is the difference between traditional programming and machine learning?

SECTION B — BASIC INTERVIEW QUESTIONS

- Q1. Why is Python used in Machine Learning?
- Q2. Name any 5 applications of AI.
- Q3. What is data in Machine Learning?
- Q4. What do you understand by training a model?
- Q5. What is the role of data in AI systems?
- Q6. What is the difference between a program and an algorithm?
- Q7. Can a machine learn without data? Why?

SECTION C — PYTHON BASICS

- Q1. What is a variable in Python? Give one example.
- Q2. What is the difference between Integer, Float, and String?
- Q3. Write the syntax of print() and input().
- Q4. Predict the output: `a = 10 b = 20 print(a + b)`
- Q5. Predict the output: `name = "Rahul" print("Hello", name)`

SECTION D — PROGRAMMING QUESTIONS

- Q1. Write a Python program to print 'Hello World'.
- Q2. Write a Python program to add two numbers using user input.
- Q3. Write a Python program to take student details and display them.
- Q4. Write a Python program to calculate the area of a circle.
- Q5. Write a Python program to convert Celsius to Fahrenheit.
- Q6. Write a Python program to create a simple calculator.

SECTION E — THINKING QUESTIONS

- Q1. If YouTube recommends videos to users, which field is being used? Explain your answer.
- Q2. Suppose a teacher groups students based on marks without labels. Which type of learning is this? Why? **BONUS QUESTION**

- Q1. Research and write about: 'How does Instagram detect faces in photos?'

MINI CHALLENGE

- Q1. Write a program that asks the user to enter their name and prints: Welcome to AI Learning